

HDOC 37

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Batch: 16

1. Algorithm

1. Start main program.
2. Display the menu with options 1 to 4.
3. Read choice and clear the input buffer.
4. Case 1 (Uppercase Conversion): Read string. Iterate through characters. If a character is between 'a' and 'z', subtract 32 from its ASCII value to convert it to uppercase. Print the modified string.
5. Case 2 (Reverse String): Read string. Calculate its length using a loop. Loop backward from length - 1 down to 0 and print each character.
6. Case 3 (Palindrome Check): Read string. Find length. Loop from $i = 0$ to $\text{length} / 2$. If $\text{str}[i] \neq \text{str}[\text{length} - 1 - i]$, it is not a palindrome. Otherwise, it is. Print result.
7. Case 4 (Exit): Terminate program.
8. Stop

2. Test-Case Table

TC ID	Scenario	Inputs	Expected Output
TC-01	Lowercase to Uppercase	Choice: 1, String: "hello 123"	Uppercase string: HELLO 123
TC-02	Reverse String	Choice: 2, String: "world"	Reversed string: dlrow
TC-03	Palindrome Check (Valid)	Choice: 3, String: "madam"	The string is a palindrome.
TC-04	Palindrome Check (Invalid)	Choice: 3, String: "hello"	The string is not a palindrome.
TC-05	Exit	Choice: 4	Exiting program...

3. C Program Source Code

```
#include <stdio.h>

void toUpperCase(void); void
reverseString(void); void
checkPalindrome(void);
void getString(char str[], int size);

int main() {
    int choice; do {
        printf("\n--- String Operations Menu 2 ---\n"); printf("1. Convert to uppercase\n");
        printf("2. Reverse a string\n"); printf("3. Check if
        palindrome\n"); printf("4. Exit\n");
        printf("Enter your choice: "); scanf("%d", &choice);

        while (getchar() != '\n');

        switch (choice) { case 1:
            toUpperCase(); break;
            case 2:
                reverseString(); break;
            case 3:
                checkPalindrome(); break;
            case 4:
                printf("Exiting program...\n"); break;
            default:
                printf("Error: Invalid choice.\n");
        }
    } while (choice != 4);

    return 0;
}

void getString(char str[], int size) { printf("Enter a string: ");
    fgets(str, size, stdin);
    int i = 0;
    while (str[i] != '\0') {
        if (str[i] == '\n') {
            str[i] = '\0'; break;
        } i++;
    }
}
```

```

void toUpperCase(void) { char str[100];
    getString(str,100); int i = 0;
    while (str[i] != '\0') {
        if (str[i] >= 'a' && str[i] <= 'z') { str[i] = str[i] - 32;
        } i++;
    }
    printf("Uppercase string: %s\n", str);
}

```

```

void reverseString(void) { char
    str[100]; getString(str, 100); int
    len = 0;
    while (str[len] != '\0') { len++;
    }
    printf("Reversed string: ");
    for (int i = len - 1; i >= 0; i--) {
        printf("%c", str[i]);
    }
    printf("\n");
}

```

```

void checkPalindrome(void) { char str[100];
    getString(str, 100);
    int len = 0;
    while (str[len] != '\0') { len++;
    }
    int isPalindrome = 1;
    for (int i = 0; i < len / 2; i++) { if (str[i] != str[len - 1 -
        i]) {
        isPalindrome = 0; break;
    }
    }
    if (isPalindrome) {
        printf("The string is a palindrome.\n");
    } else {
        printf("The string is not a palindrome.\n");
    }
}

```

5. Compilation and Execution

```
(kali@kali)-[~/liak]
$ ./a.out

— String Operations Menu 2 —
1. Convert to uppercase
2. Reverse a string
3. Check if palindrome
4. Exit
Enter your choice: 1
Enter a string: Keshav HDOC repo
Uppercase string: KESHAV HDOC REPO

— String Operations Menu 2 —
1. Convert to uppercase
2. Reverse a string
3. Check if palindrome
4. Exit
Enter your choice: 2
Enter a string: Keshav HDOC repo
Reversed string: oper CODH vahseK

— String Operations Menu 2 —
1. Convert to uppercase
2. Reverse a string
3. Check if palindrome
4. Exit
Enter your choice: 3
Enter a string: helleh
The string is a palindrome.

— String Operations Menu 2 —
1. Convert to uppercase
2. Reverse a string
3. Check if palindrome
4. Exit
Enter your choice: 3
Enter a string: Palindrome
The string is not a palindrome.

— String Operations Menu 2 —
1. Convert to uppercase
2. Reverse a string
3. Check if palindrome
4. Exit
Enter your choice: 4
Exiting program...
```